

SUPPLEMENTARY MATERIAL

Deposition and Characterization of PVD Thin Films

4th Year Integrated MSc. Open Lab Report

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Procedure

Operation of the PVD System

1. The PVD system was initially kept in the **close position** (all the knobs) before starting the operation.
2. The **chiller** was switched on by first turning on the main supply and subsequently switching on the compressor.
3. The **rotary pump** was switched on using the rotary-pump control knob to establish the initial vacuum in the chamber, just after shifting the lever to the backing position.
4. The backing valve/lever was kept in the appropriate **backing position** during the initial pumping stage.
5. The rotary pump was allowed to operate until the required backing pressure was achieved. In the present procedure, the backing pressure was allowed to reach approximately

$$P_{\text{backing}} \approx 10^{-2} \text{ mbar.} \quad (1)$$

6. After achieving the required backing pressure, the lever was shifted to the roughing position, and both the backing and roughing pressures were checked to be of the order of 10^{-2} mbar.
7. After achieving the required pressure on the Pirani gauge, the **diffusion pump** was started by keeping the lever in the roughing position.
8. The system was allowed to operate for approximately **45 minutes** so that the diffusion pump could reach its proper operating condition.
9. The **high-vacuum valve** was then opened gradually to establish high vacuum in the deposition chamber.
10. The chamber pressure was allowed to decrease until a high-vacuum condition of approximately

$$P_{\text{chamber}} < 5 \times 10^{-5} \text{ mbar} \quad (2)$$

was reached, as read on the Penning gauge.

11. After obtaining the required vacuum, the substrate was positioned appropriately inside the deposition chamber by admitting air into the deposition chamber, before closing the high-vacuum valve and setting the Pirani gauge lever to the backing position.
12. The material to be deposited was loaded into the target/sample holder (boat/bucket), after first being weighed.
13. After closing the chamber door, it was ensured that the air-admittance knob was closed; the high-vacuum valve was then reopened, with the Pirani gauge lever kept in the backing position.
14. The LT (low-tension) current was switched on and gradually increased until proper vaporisation of the source material was observed through the viewer glass.
15. After allowing some time for cooling, the high-vacuum valve was closed, the air-admittance valve was opened, and the chamber was then opened.
16. The deposited samples were removed from the chamber after completion of the deposition process and were retained for subsequent characterization.

Shutdown Procedure

17. After completion of the deposition, the **high-vacuum valve** was closed.
18. The diffusion pump was switched off using its corresponding control.
19. The system was returned to the **backing position** to maintain the appropriate pumping configuration during cooling.
20. The system was allowed to cool for approximately **30 minutes** before proceeding with further shutdown operations.
21. The rotary pump was then switched off.
22. The chiller was switched off after ensuring that the system had undergone the required cooling period.
23. Finally, the **main power supply** was switched off.

Note : *The above mentioned is for the set-up model BC-100, there are some limitations as per actual use but if you want original manual procedure then visit this [link](#)*

What We Did Differently / Survival Guide

Note: We made the mistakes so you don't have to. Read this before you touch anything and strictly follow it.

- **Handle Substrates Cleanly (and Mind the Diamond Cutter):** Always clean your substrates thoroughly before mounting. When scoring and snapping glass or silicon with the diamond scribe, use light, consistent pressure—work carefully so you don't end up with uneven edges or accidental cuts.
- **Keep Bare Hands Out of the Chamber:** Never touch substrates, target materials, or internal chamber fixtures with bare fingers. Fingerprint oils outgas rapidly under high vacuum, killing your base pressure and ruining film adhesion. Always wear clean nitrile gloves.
- **Maintain a Clutter-Free Zone:** Never drop or store flammable items beneath the setup. The heated diffusion pump at the bottom reaches temperatures high enough to instantly ignite paper, wipes, or solvents on contact.
- **Chiller On First (And Visually Confirm Flow):** Turn on the water chiller before starting any vacuum pumping. Don't just trust the power light—pop open the chiller box and visually verify that cooling water is actively flowing through the hose connected to the diffusion pump jacket.
- **Level and Torque the Target Holder Evenly:** When securing your boat/target holder, tighten both sides evenly to the exact same level. Uneven clamping creates localized electrical resistance, which causes heavy sparking and will instantly snap your holder mid-run.
- **Line-of-Sight Substrate Alignment:** Position your substrate on the sample disk directly in the line of sight of the target, keeping solid-angle deviation to a minimum. Off-center placement guarantees an uneven, non-uniform film thickness.
- **Follow the Valve Sequence Religiously:** Execute every step of the pumping sequence (Backing → Roughing → High Vacuum) in exact order as mentioned in procedure in detail. Bypassing or swapping steps will stall the system, leaving you wondering why the chamber won't pump down.
- **Hold Off on the High-Vac Valve:** Never open the high-vacuum (gate) valve until the roughing/backing stage has pulled the vacuum down to its required pressure threshold. Opening it too early backstreams oil and halts high-vacuum operation forever.
- **Beware of Ghost Readings on the Pirani Gauge:** If the pressure stalls during backing or roughing, check the Pirani gauge sensor on the back of the machine. Rotary pump oil can occasionally backstream into the gauge head, coating the filament and causing completely inaccurate pressure readings.
- **Ramp the LT Current Slowly:** Once the source material begins to melt, increase the Low-Tension (LT) current very gradually over at least 5 minutes. Rushing the current causes thermal shock, violent spitting of the source material, and poor film quality.

- **Avoid the “Fresh Deposition” Fumes:** When venting and opening the chamber right after a run, step back and let the chamber air out. Wear a mask if possible to avoid directly breathing in fine thermal vapor particulates or hot chemical odors.
- **Never Pull the Plug on a Hot Diffusion Pump:** Never shut off the power directly! Turn off the diffusion pump heater, keep the system in backing mode, and let the pump cool down for at least 30 minutes. Once cooled, complete the shutdown sequence step-by-step in reverse order (Rotary Pump → Chiller → Main Power) detailed mention in procedure section.

Note: I feel you are thinking, “ahh let it be, we will see later.” Okk rules are meant to be broken but I think after checking [this link](#), you will regret not reading it properly. Still, you have time to do so - a few minutes might save you a month.

To The Future Researchers & Students: Consider this survival guide a living document! If you discover new quirks, troubleshooting tricks, or practical tips during your runs, please update this list and cite this report in your work ([Git-Repo](#)). Passing down hard-earned practical knowledge ensures the next batch of students can master the setup safely, avoid costly mistakes, and focus on getting good data.

Images

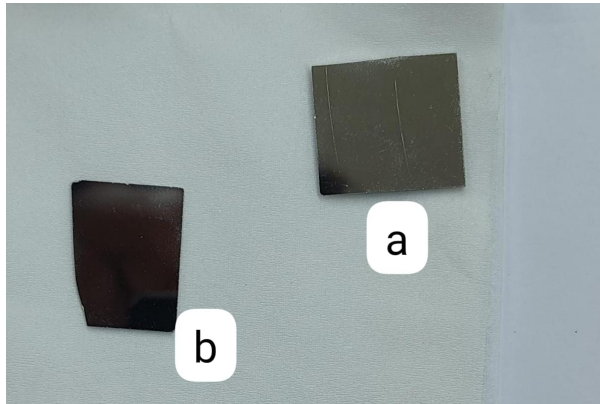


Figure 1: Aluminum thin films deposited on glass substrates.

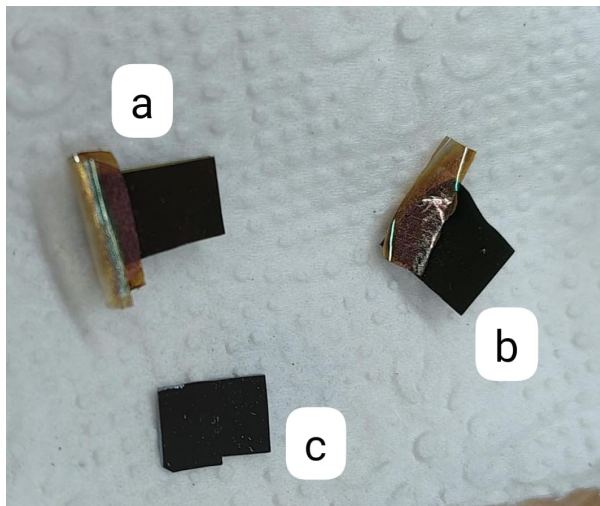


Figure 2: Silicon thin-film samples: (a,b) p-type Si thin films deposited on p-type Si substrates, removable cello tape for masking, and (c) pristine p-type Si substrate.

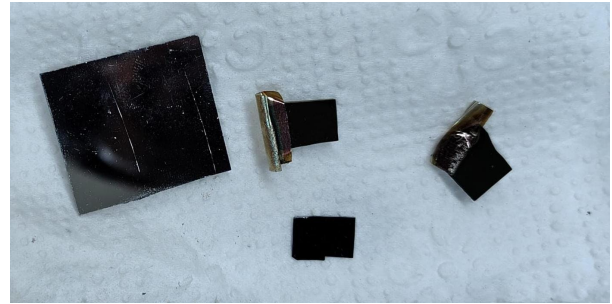


Figure 3: Overview of all prepared samples, including Al thin films on glass, p-type Si on p-type Si, and the unmodified p-type Si substrate.

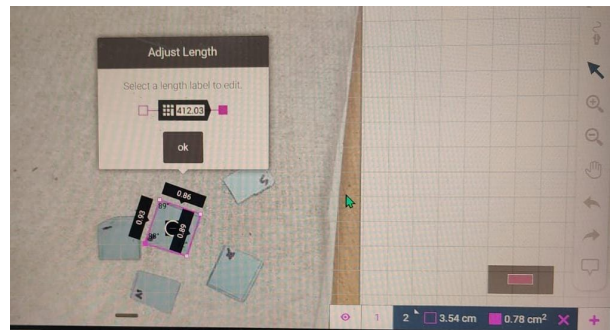


Figure 4: Surface area calculation depiction using image and calc software, by marking each substrate.



Figure 5: Ultrasonic cleaning process.



Figure 6: Substrate attaching to circular plate inside vacuum chamber.



Figure 7: substrate marked and weighted as in ref. table

Sample No.	Mass (g)	Surface Area (cm ²)
1	0.063	—
2	0.068	—
3	0.078	0.78
4	0.078	0.78
5	0.073	—
6	0.069	—